

Lab09Report

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```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggrepel)
library(corrplot)
```

corrplot 0.95 loaded

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

filter

The following object is masked from 'package:graphics':

layout

```
library(skimr)

candy_file <- "candy-data.csv"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

There are 85 candy types

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

There are 38 fruity candies

```
sum(candy$fruity)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) and its winpercent?

My favorite candy is Snickers and its winpercent is 76.67

```
candy["Snickers", "winpercent"]
```

```
[1] 76.67378
```

Q4. What is the winpercent for Kit Kat?

The winpercent is 76.77 for kit kat.

```
candy["Kit Kat", "winpercent"]
```

```
[1] 76.7686
```

Q5. What is the winpercent for Tootsie Roll Snack Bars?

The winpercent is 49.65 for Tootsie Rolls

```
candy["Tootsie Roll Snack Bars", "winpercent"]
```

```
[1] 49.6535
```

Q6. Is any variable on a different scale?

Yes. winpercent is on a different scale (0–100), while most other variables are binary (0 or 1).

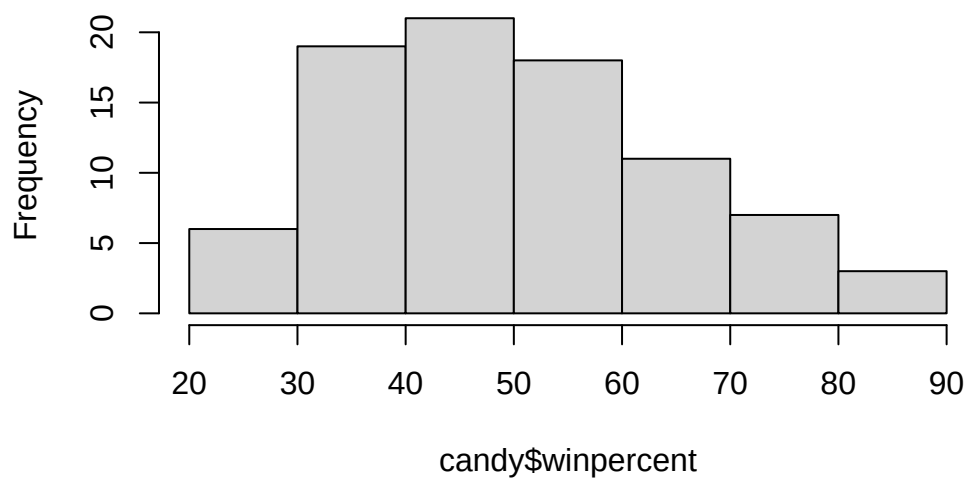
Q7. What do 0 and 1 represent in chocolate?

1 = contains chocolate 0 = does not contain chocolate

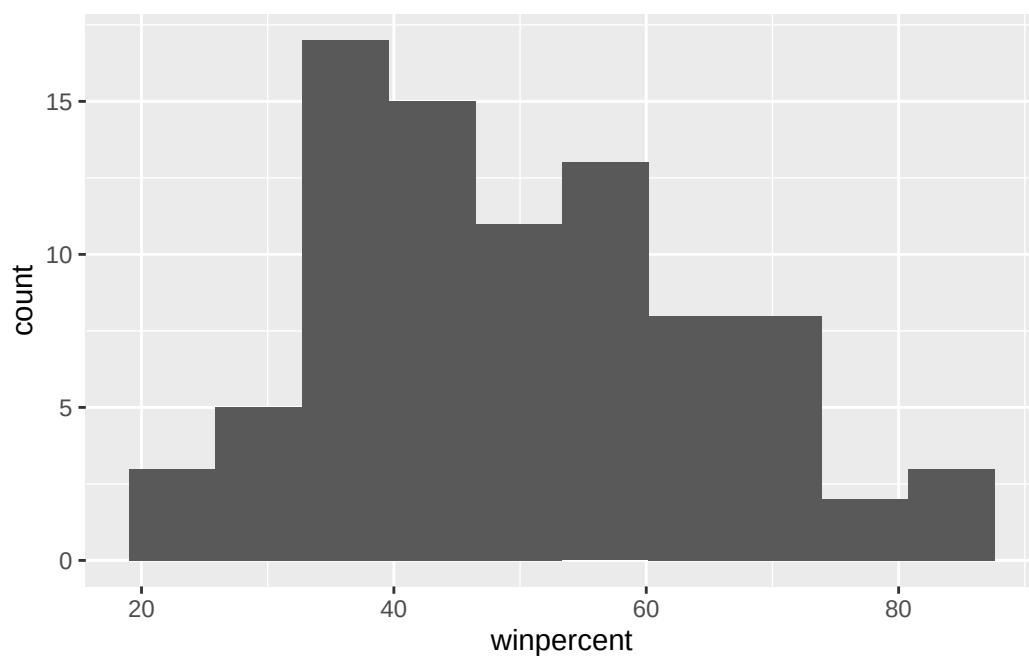
Q8. Plot a histogram of winpercent

```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
ggplot(candy, aes(x = winpercent)) +  
  geom_histogram(bins = 10)
```



Q9. Is the distribution symmetrical? No, it is slightly skewed to the left.

Q10. Is the center above or below 50%? Slightly below 50%.

Q11. Is chocolate or fruity candy ranked higher? Chocolate candy is ranked higher on average.

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant? Yes, it is statistically significant.

```
t.test(candy$winpercent[candy$chocolate == 1],  
       candy$winpercent[candy$fruity == 1])
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Q13. What are the five least liked candies? Nik L Nip Boston Baked Beans Chiclets Super Bubble Jawbusters

```
head(candy[order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0

Jawbusters	0	1	0	0	0	
	crisped	ricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip		0	0	0	1	0.197
Boston Baked Beans		0	0	0	1	0.313
Chiclets		0	0	0	1	0.046
Super Bubble		0	0	0	0	0.162
Jawbusters		0	1	0	1	0.093
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

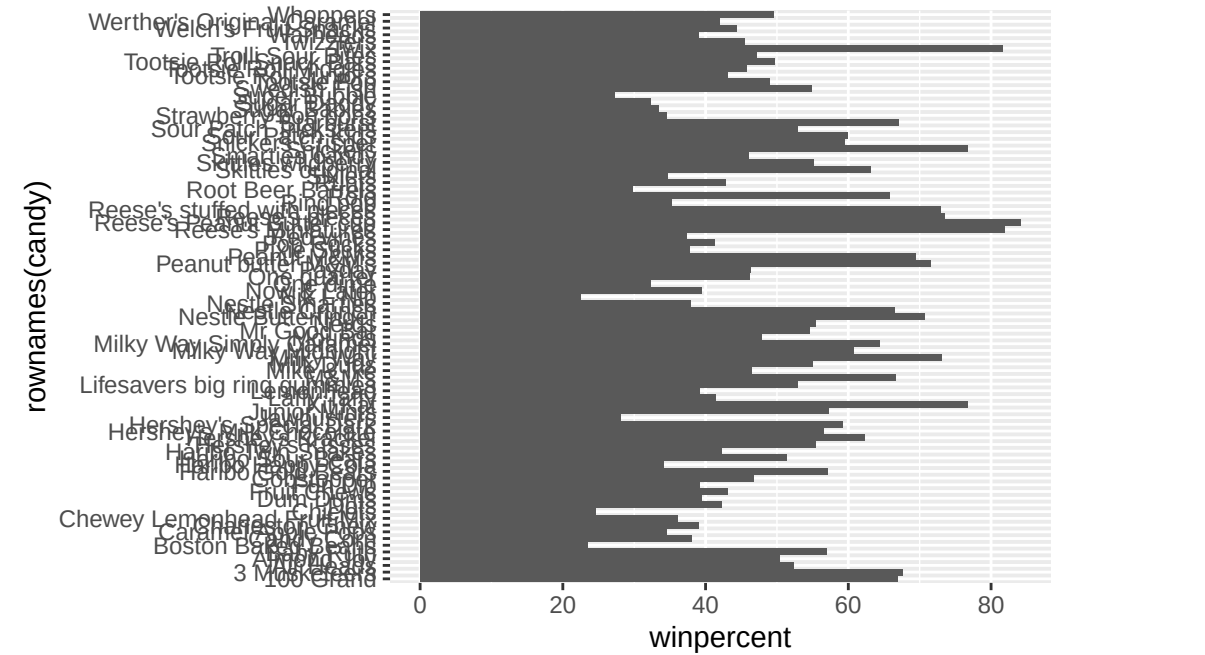
Q14. What are the top 5 favorite candies? Reese's Peanut Butter Cup Reese's Miniatures
Twix Kit Kat Snickers

```
head(candy[order(candy$winpercent, decreasing = TRUE), ], 5)
```

	chocolate	fruity	caramel	peanut	almondy	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1
	crisped	ricewafer	hard	bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0	0.720
Reese's Miniatures		0	0	0	0	0.034
Twix		1	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Snickers		0	0	1	0	0.546
	pricepercent	winpercent				
Reese's Peanut Butter cup	0.651	84.18029				
Reese's Miniatures	0.279	81.86626				
Twix	0.906	81.64291				
Kit Kat	0.511	76.76860				
Snickers	0.651	76.67378				

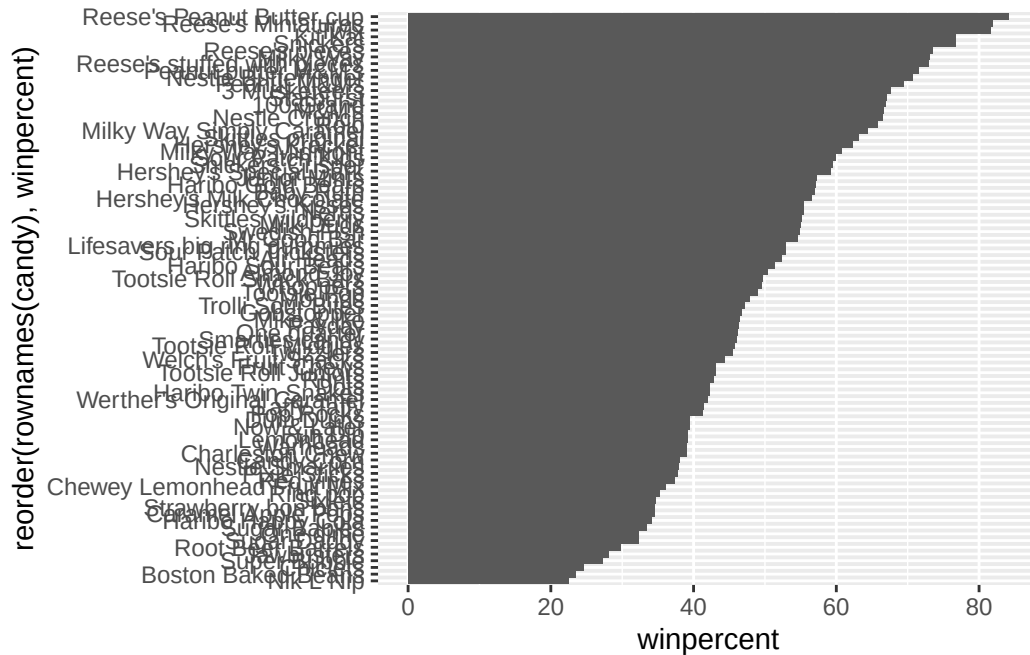
Q15. Make a barplot of rankings

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



Q16. Improve with sorting

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



Q17. Worst ranked chocolate candy? Sixlets

Q18. Best ranked fruity candy? Starburst

Q19. Best candy for lowest price (“bang for buck”)? Reese’s Miniatures

Q20. Top 5 most expensive candies & least popular among them Most expensive:
Nik L Nip Nestle Smarties Ring Pop Hershey’s Krackel Hershey’s Milk Chocolate

Least popular: Nik L Nip

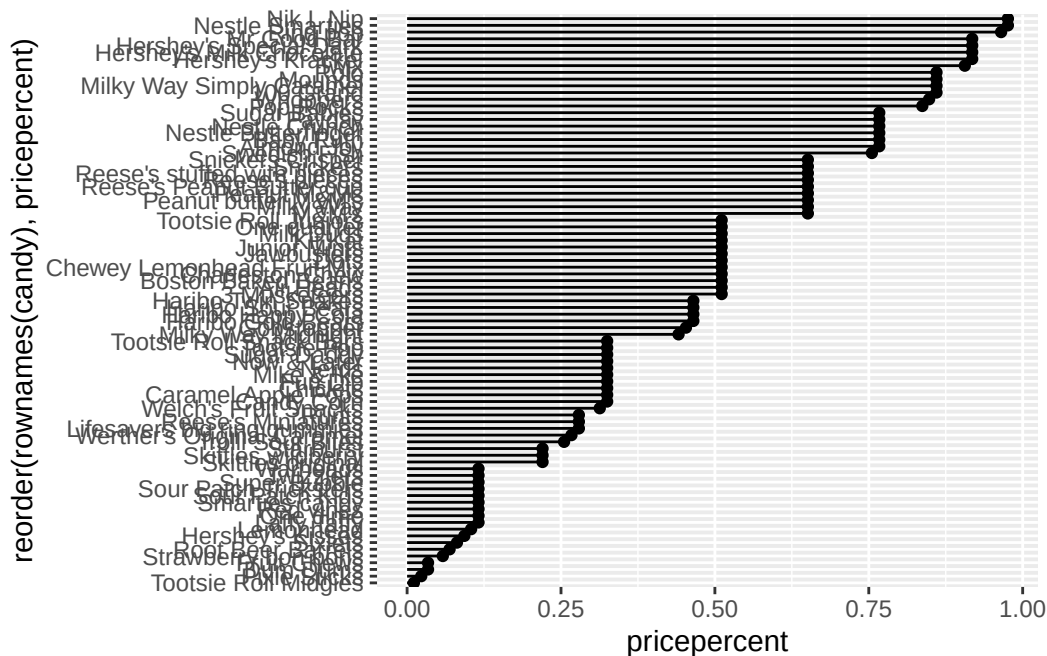
```
head(candy[order(candy$pricepercent, decreasing = TRUE), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Nestle Smarties	1	0	0		0	0
Ring pop	0	1	0		0	0
Hershey's Krackel	1	0	0		0	0
Hershey's Milk Chocolate	1	0	0		0	0
	crisped	rice	wafer	hard bar	pluribus	sugarpercent
Nik L Nip		0	0	0	1	0.197
Nestle Smarties		0	0	0	1	0.267
Ring pop		0	1	0	0	0.732
Hershey's Krackel		1	0	1	0	0.430
Hershey's Milk Chocolate		0	0	1	0	0.430

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Q21. Lollipop chart

```
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                    xend = 0)) +
  geom_point()
```



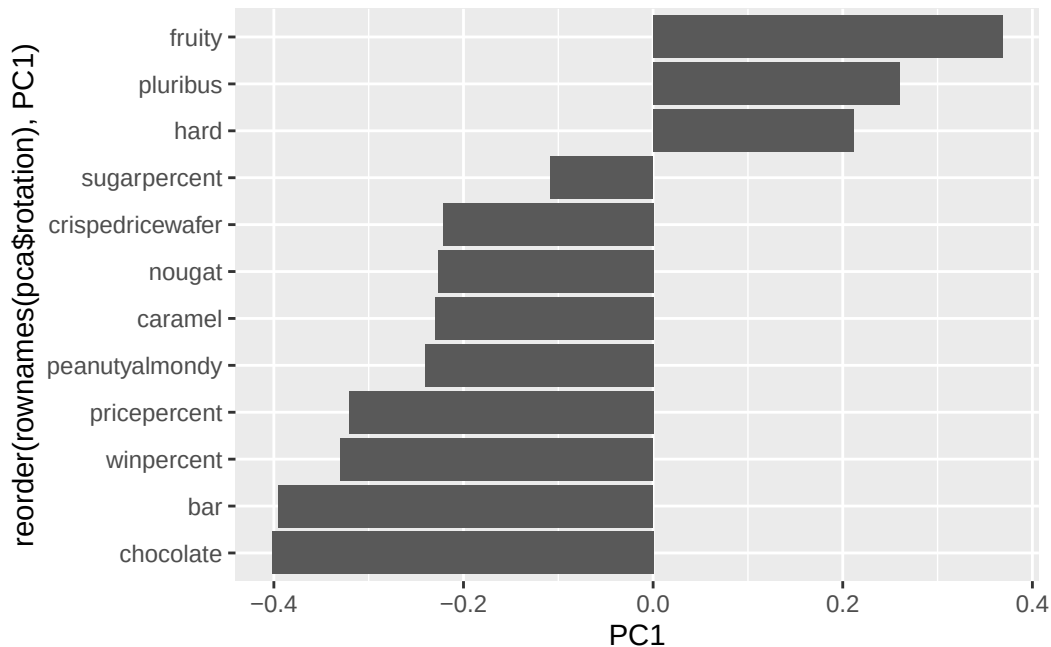
Q22. Which variables are anti-correlated? chocolate and fruity

Q23. Which variables contribute most to PC1? Chocolate (negative direction) Fruity (positive direction) Bar Winpercent

Q24. PCA loadings plot

```
pca <- prcomp(candy, scale = TRUE)

ggplot(as.data.frame(pca$rotation)) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Positive PC1: fruity, hard candy Negative PC1: chocolate, bar, winpercent

Q25. What makes a “winning” candy? A winning candy is typically:

Chocolate-based Often a candy bar Not fruity or hard candy

Bar plots show top candies are chocolate (Reese’s, Twix, Kit Kat) Correlation shows chocolate positively linked with popularity PCA shows chocolate vs fruity separation as main pattern